Niladri Talukder

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PROFILE:

Ph.D. Mechanical Engineer with strong expertise in advanced materials characterization, structure–property–performance analysis, and defect/degradation evaluation of functional nanomaterials for electrochemical and thermal energy systems. Experienced in SEM, TEM/STEM, EDS, XPS, Raman, FTIR, XRD, DSC, TGA, electrochemical testing, and complex data interpretation to correlate microstructural, chemical, and performance changes in solid materials. Proven record of peer-reviewed publications, independent research execution, student mentoring, external collaboration with national laboratories, and technical communication through papers, proposals, reports, and presentations.

SKILLS:

- **Advanced Materials Characterization**: SEM, TEM, STEM, EDS, XPS, Raman spectroscopy, FTIR, XRD, DSC, TGA, and electrochemical characterization for microstructural, chemical, thermal, surface, and structure–property analysis.
- **Electron Microscopy & Degradation Analysis**: Electron microscopy-based evaluation of nanomaterials, electrode materials, structural evolution/degradation of materials, and related performance changes of the materials.
- **Structure–Property–Performance Correlation**: Correlating processing conditions, microstructure, defects, degradation pathways, and functional performance in nanomaterials, composites, catalysts, and energy materials.
- **Materials Synthesis & Processing:** Graphene, N-doped graphene, MOFs, MXenes, carbon-based nanocomposites, and functional electrode materials; ball milling, ultrasonication, thermal treatment, and chemical processes.
- **Experimental Design & Workflow Development**: Independent design and execution of controlled experiments, reproducibility validation, sample preparation, characterization workflow development, troubleshooting, and documentation of experimental procedures, training research assistants on the experimental procedures.
- **Data Analysis, Visualization & Technical Computing**: Scientific data interpretation, statistical analysis, visualization of complex datasets, and Python/MATLAB-assisted data processing for research analysis and reporting.
- **Research Communication & Scientific Writing**: Peer-reviewed journal publications, conference presentations, technical reports, research proposals, and communication of complex results to multidisciplinary audiences.
- **Mentoring & Collaborative Research**: Coordinating concurrent research activities, collaborating with national laboratory scientists and external industry partners, and supporting multidisciplinary project execution.

EDUCATION:

Degree	***Institution***	***Duration***
Ph.D. in Mechanical Engineering	New Jersey Institute of Technology, United States	Sep 2020 — Dec 2025
M.Sc. in Mechanical Engineering	Andong National University, South Korea	Sep 2015 — Aug 2017
B.Sc. in Mechanical Engineering	Khulna University of Eng. & Tech., Bangladesh	Jan 2009 — Sep 2013

PROFESSIONAL EXPERIENCES:

Abonics, Inc., NJ, USA. [*a Startup: spin-off from New Jersey Institute of Technology*]

Position: Chief Scientist *May 2026 — Current*

- Lead the company's technology strategy, research, and product development for biochip platforms, including design, optimization, validation, and the transition from prototype to scalable solutions aligned with clinical and market needs.
- Build and lead a multidisciplinary technical team to materialize innovations, secure the intellectual properties. Ensure compliance with relevant regulatory and quality standards to support product reliability and clinical readiness.

New Jersey Institute of Technology (NJIT), NJ, USA.

Position: Research/Teaching Assistant Sep 2020 — Jan 2026

- Project Lead: ***Carbon-based nanomaterials for electrochemical energy conversion and storage systems.***
  - Synthesize nitrogen-doped graphene/ZIF-8 (N-G/MOF) nanocatalysts for the oxygen reduction reaction (ORR) via ball milling and thermal treatment; optimize N-doping ratio and active sites by controlling milling speed and time, achieving catalytic performance surpassing N-G and approaching benchmark 10 wt% Pt/C catalyst performance.

- Evaluate long-term stability and degradation mechanisms of N-G/MOF nanocatalysts under ORR conditions using electrochemical and advanced structural characterization techniques, establishing relationships between physical and chemical structures and catalytic performance under different operational environments.

➢ Project Lead: ***Nanomaterials-supported Phase Change Materials (PCMs) for thermal property enhancement.***
- Leverage molecular-level interactions of engineered nanomaterials and PCMs to enhance thermal performance. Achieved ~5% increase in latent heat and ~10% improvement in thermal stability relative to the base PCM.
- Lead a three-member team to assess the commercialization potential of Nano-PCM composites through the NSF I-Corps program. Conducted customer discovery interviews for market analysis, and value proposition validation.

➢ Project Co-lead: ***Microfluidic-based multiplex assay biochip development for point-of-care disease detection.***
- Developed capacitance-based biosensors integrated with microfluidic biochips for rapid detection of ovarian cancer biomarkers CA-125 and HE4 and Alzheimer's disease biomarkers Aβ42/Aβ40. Achieved biomarker detection in less than 40 seconds with high sensitivity and specificity. Ran commercialization efforts with this technology.

➢ Maintained safe laboratory operations, including equipment oversight and raw material procurement, ensure regulatory compliance and support reliable prototype development and validation.

➢ Mentored 7+ undergraduate and graduate researcher assistants in materials synthesis, electrochemical testing, and data analysis while coordinating concurrent experimental activities and ensuring timely technical outcomes.

➢ Served as Teaching Assistant for 12 undergraduate and graduate engineering courses at NJIT for 4 years; maintained clear technical communication, delivered lectures and laboratory instructions, monitored students' progress.

Center for Functional Nanomaterials (CFN), Brookhaven National Laboratory, NY, USA.

Role: Visiting Researcher [*through general user proposals*] *May 2021 — Dec 2025*

➢ Led four experimental research projects on the development of functional nanomaterials, emphasizing structure–property relationships for electrochemical fuel cells, batteries, and thermal energy storage materials.

➢ Established and maintained collaborations with scientists at the Center for Functional Nanomaterials (CFN); enabled advanced characterization and high-impact investigations of electrode materials and degradation mechanisms.

Source Associate Ltd., Bangladesh.

Position: Executive (Engineering). *Oct 2019 — Dec 2020*

➢ Oversaw plant erection and installation activities to ensure compliance with NFPA standards; managed material planning and control processes; and developed detailed project plans and Bills of Quantities (BOQ).

Andong National University, South Korea.

Position: Graduate Research Assistant. *Sep 2015 — Oct 2017*

➢ Led a project for designing and operating a constant-volume combustion chamber with Schlieren and shadowgraph diagnostics for high-precision combustion studies. Measured laminar flame speed and Markstein length.

SELECTED PUBLICATIONS:

Peer-reviewed Journal Articles: (3 out of 15)

1. **Talukder, N.**; Wang, Y.; Tong, X.; Lee, E. S. Chemical Changes from N-doped Graphene and Metal-organic Frameworks to N-G/MOF Composites for Improved Electrocatalytic Activity. *Carbon* **2025**, 232, 119816. [Link]
2. **Talukder, N.**; Wang, Y.; Nunna, B. B.; Tong, X.; Lee, E. S. An Investigation on the Structural Stability of ZIF-8 in Water versus Water-derived Oxidative Species in Aqueous Environment. *Micropor. Mesopor. Mater.* **2024**, 366, 112934. [Link]
3. **Talukder, N.**; Wang, Y.; Nunna, B. B.; Tong, X.; Boscoboinik, A. J.; Lee, E. S. Investigation on Electrocatalytic Performance and Material Degradation of N-doped Graphene-MOF Nanocatalyst in Emulated Electrochemical Environments. *Industrial Chemistry & Materials* **2023**, 1, 360−375. [Link]

ACHIEVEMENTS, and AWARDS:

➢ Published 17 research articles in reputed scientific journals, international conferences (*Google Scholar Link*).

➢ Authored a book on Nitrogen-doped Graphene Nanomaterials (13 Chapters); by *Springer Nature*. *[Link]*

➢ One non-provisional Patent Application through NJIT on nanomaterials supported Phase Change Materials. [*Link*]

➢ Five National Science Foundation (NSF) I-Corps grant awards for research and technology commercialization.